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GNIOT Institute of Management Studies – PGDM (Batch 2022-24)
(TRIMESTER – II) END TERM EXAMINATION, DECEMBER- 2022
OPERATIONS MANAGEMENT

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A**(Short Answer Type Questions, do not exceed 100 words)****Q.1. Attempt all parts of the following:****(10×3=30)**

- (a) Illustrate the importance of location for setting up a petrol pump.
- (b) Is your sales team effective or efficient if all the targeted visits have been made but the targeted orders could not be achieved in the last month? Explain the concept.
- (c) Distinguish between method study and work measurement.
- (d) Identify the types of inventories.
- (e) Explain total productive maintenance with examples.
- (f) Discuss the relationship between Kanban and JIT.
- (g) Differentiate between responsiveness and efficiency in supply chain management.
- (h) “Run it till it breaks.” If you have been advised to follow this principle, what maintenance mode would you suggest and why?
- (i) Explain the key concepts of scientific management.
- (j) Describe TQM with examples.

SECTION-B**(Long Answer Type Questions)****Attempt all questions of the following:****(4×10=40)**

- Q.2.(a)** Why is layout planning important? Describe the different types of layouts. Which layout would you recommend for a car manufacturing plant?

OR

- (b) When is production control exercised? Illustrate production, planning and control for a garment fabricator.

- Q.3. (a)** Explain quality with an appropriate example. You are contemplating to buy MG Hector SUV, the latest model. Evaluate the vehicle on all the service quality dimensions.

OR

- (b) ABB is a global leader in power and automation technologies. It stocks a certain switch connector at its central warehouse for supplying to its field service offices in India. The yearly demand for these connectors is 15,000 units. ABB estimates its holding cost for this item to be Rs 1500 per unit. The cost to place and process an order from the supplier is Rs 4500. The company operates 300 days per year and the lead time to receive an order from the supplier is 30 working days. Determine the optimum quantity and expected time between orders.

- Q.4.(a)** “Carry the maintenance based on actual conditions of the machine.” If you have been advised to follow this principle, what maintenance mode would you suggest and why? What are the advantages and disadvantages of this maintenance mode?

OR

- (b) (i) Discuss the basic steps of time study.
(ii) Assuming that the total observed time for an operation of assembling an electric switch is one minute, and if the rating is 120%, find normal time. If an allowance of 10% is allowed for the operation, determine the standard time.

- Q.5. (a)** We are now entering the era of “supply chain competition.” The fundamental difference from the previous model of competition is that an organisation can no longer act as an isolated and independent entity in competition. Instead, the need is to create value delivery systems that are more responsive to fast-changing markets and are much more consistent and reliable in the delivery of that value. Describe the new rules of competition that have compelled the organisations to respond to these changes.

OR

(b) For a project, several activities and their duration are given below:

Activities	Duration (Days)
1–2	4
1–3	3
1–4	6
2–4	4
2–5	8
3–4	4
3–5	4
4–5	6

Draw the network, identify critical path, and determine project completion time.

SECTION-C

Q.6. Attempt all parts of the following:

(2×15=30)

Sheena, the Entrepreneur – A Vignette

(Excerpted from “Production/Operations Management” by William J. Stevenson, Irwin/McGraw-Hill)

Sheena had worked for the same Fortune 500 Company for almost 15 years. Although the company had gone through some tough times, things were starting to turn around. Customer orders were up, and quality and productivity had improved dramatically from what they had been only a few years earlier due to companywide quality improvement program. So, it comes as a real shock to Sheena and about 400 of her co-workers when they were suddenly terminated following the new CEO’s decision to downsize the company.

After recovering from the initial shock, Sheena tried to find employment elsewhere. Despite her efforts, after eight months of searching she was no closer to finding a job than the day she started. Her funds were being depleted and she was getting more discouraged. There was one bright spot, though, she was able to bring in a little money by mowing lawns for her neighbours. She got involved quite by chance when she heard one neighbour remark that now that his children were on their own, nobody was around to cut the grass. Almost jokingly, Sheena asked him how much he would be willing to pay.

Soon Sheena was mowing the lawns of five neighbours. Other neighbours wanted her to work on their lawns, but she did not feel that she could spare any more time from her job search. However, as the rejection letters began to pile up, Sheena knew she had to make an important decision in her life. On a rainy Tuesday morning, she decided to go into business for herself taking care of neighbourhood lawns. She was relieved to give up the stress of job hunting, and she was excited about the prospects of being her own boss. But she was also fearful of being completely on her own. Nevertheless, Sheena was determined to make a go of it.

At first, business was a little slow, but once people realized Sheena was available, many asked her to take care of their lawns. Some people were simply glad to turn the work over to her; others switched from professional lawn care services. By the end of her first year in business, Sheena knew she could earn a living this way. She also performed other services such as fertilizing lawns, weeding gardens, and trimming shrubbery. Business became so good that Sheena hired two part-time workers to assist her and, even then, she believed she could expand further if she wanted to.

Q6 (a) Sheena is the operations manager of her business. Among her responsibilities are forecasting, inventory management, scheduling, quality assurance, and maintenance.

- What kinds of things would likely require forecasts?
- What inventory items does Sheena probably have?
- Name one inventory decision she has to make periodically.
- What things might occur to disrupt schedules and cause Sheena to reschedule?
- What kinds of maintenance must be performed?

Q6 (b) The town is considering an ordinance that would prohibit putting grass clippings at the curb for pickup because local landfills cannot handle the volume.

- What options might Sheena consider if the ordinance is passed?
- Name the advantages and drawbacks of each option.